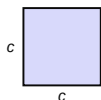


Geometry formula sheet · 4t ESO

Plane figures: area and perimeter

A : area · P : perimeter (the sum of all the sides).

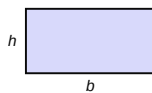
Square



$$A = c^2$$

$$P = 4c$$

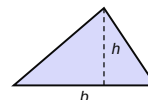
Rectangle



$$A = b \cdot h$$

$$P = 2b + 2h$$

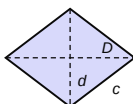
Triangle



$$A = \frac{b \cdot h}{2}$$

$$P = \text{sum of the 3 sides}$$

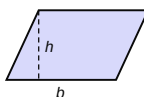
Rhombus



$$A = \frac{D \cdot d}{2}$$

$$P = 4c$$

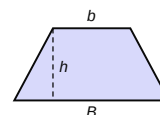
Parallelogram



$$A = b \cdot h$$

$$P = \text{sum of the sides}$$

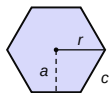
Trapezium



$$A = \frac{(B + b) \cdot h}{2}$$

$$P = \text{sum of the sides}$$

Regular polygon

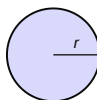


$$P = n \cdot c$$

$$A = \frac{P \cdot a}{2}$$

n : number of sides · a : apothem · r : radius. In a **regular hexagon** the side equals the radius, $c = r$, and the apothem comes from Pythagoras: $a^2 = r^2 - \left(\frac{c}{2}\right)^2$.

Circle

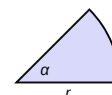


$$A = \pi r^2$$

$$L = 2\pi r$$

$$L: \text{length of the circumference}$$

Circular sector

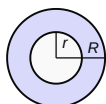


$$A = \frac{\pi r^2 \cdot \alpha}{360^\circ}$$

$$L_{\text{arc}} = \frac{2\pi r \cdot \alpha}{360^\circ}$$

$$P = L_{\text{arc}} + 2r$$

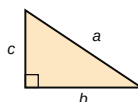
Annulus



$$A = \pi (R^2 - r^2)$$

$$P = 2\pi R + 2\pi r$$

Pythagoras' theorem



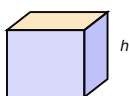
$$a^2 = b^2 + c^2$$

In a right triangle, a is the hypotenuse.

Solids: area and volume

A_B : base area · A_L : lateral area · A_T : total area · P_B : perimeter of the base · V : volume.

Prism



A_B = area of the base polygon

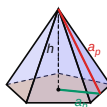
$$A_L = P_B \cdot h$$

$$A_T = 2A_B + A_L$$

$$V = A_B \cdot h$$

Cube with edge c : $A_T = 6c^2$, $V = c^3$.

Pyramid



A_B = area of the base polygon

$$A_L = n \cdot A_{\triangle}$$

$$A_L = \frac{P_B \cdot a_p}{2}$$

$$A_T = A_B + A_L$$

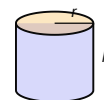
$$V = \frac{A_B \cdot h}{3}$$

The side faces are n equal triangles. **1st way:**

$A_L = n \cdot A_{\triangle}$, with $A_{\triangle} = \frac{c \cdot a_p}{2}$ (base c , side of the polygon; height a_p , slant height). **2nd way:**

since $n \cdot c = P_B$, altogether $A_L = \frac{P_B \cdot a_p}{2}$. If the base is a regular polygon, $A_B = \frac{P_B \cdot a_b}{2}$ (a_b : apothem of the base). Pythagoras: $a_p^2 = h^2 + a_b^2$.

Cylinder



$$A_B = \pi r^2$$

$$A_L = 2\pi r \cdot h$$

$$A_T = 2A_B + A_L$$

$$V = A_B \cdot h = \pi r^2 h$$

Cone



$$A_B = \pi r^2$$

$$A_L = \pi r \cdot g$$

$$A_T = A_B + A_L$$

$$V = \frac{A_B \cdot h}{3} = \frac{\pi r^2 h}{3}$$

g : slant height, $g^2 = h^2 + r^2$.

Sphere



$$A = 4\pi r^2$$

$$V = \frac{4}{3}\pi r^3$$

No base: just the total area. Hemisphere: $V = \frac{2}{3}\pi r^3$.

Capacity and units

$$1 \text{ dm}^3 = 1 \text{ L}$$

$$1 \text{ m}^3 = 1000 \text{ dm}^3 = 1000 \text{ L}$$

$$1 \text{ cm}^3 = 1 \text{ mL}$$

Areas: each unit step is $\times 100$. Volumes: each step is $\times 1000$.

Notice the difference: prisms and cylinders have **two** bases ($A_T = 2A_B + A_L$); pyramids and cones have only **one** ($A_T = A_B + A_L$), and their volume is a third of that of the prism or cylinder with the same base and height.